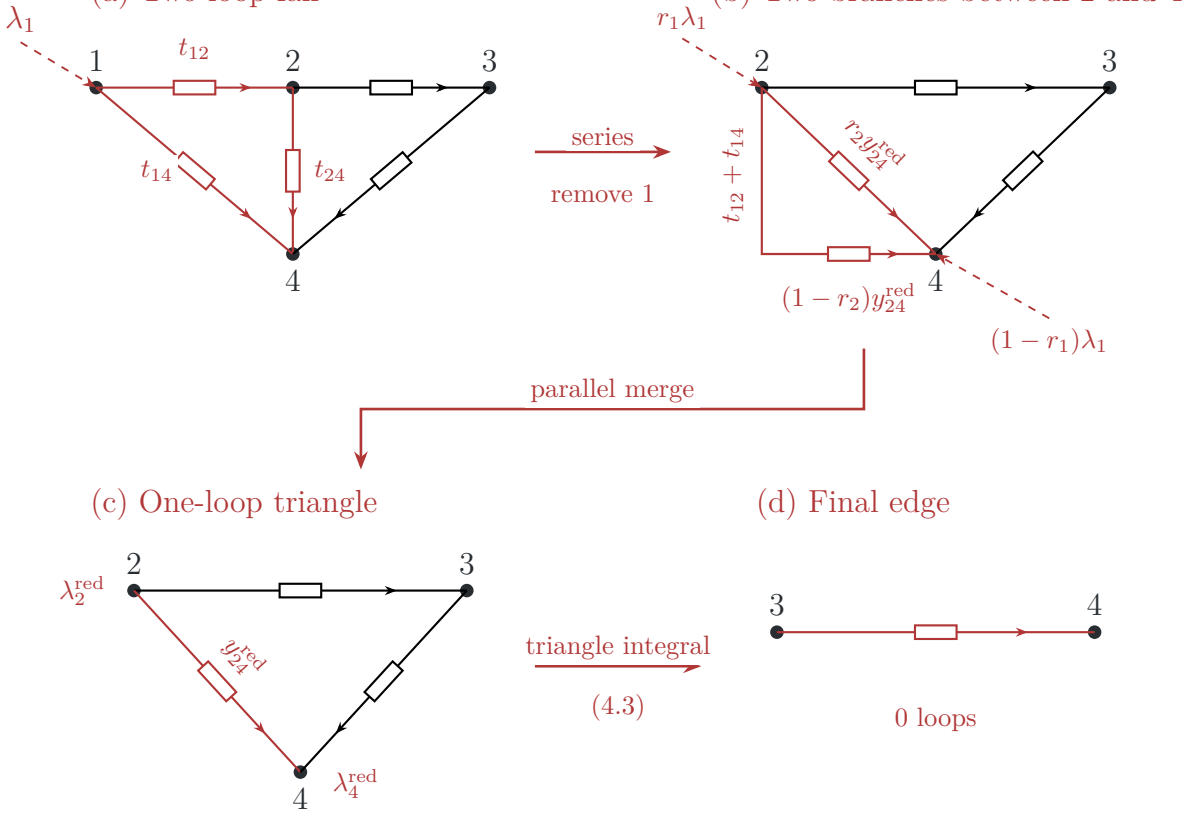

(a) Two-loop fan
λ_1
1
t_{12}
2
3
t_{14}
t_{24}
4
series
remove 1
(b) Two branches between 2 and 4
$r_1\lambda_1$
2
3
$r_2 y_{24}^{\mathrm{red}}$
$t_{12}+t_{14}$
$(1-r_2)y_{24}^{\mathrm{red}}$
4
$(1-r_1)\lambda_1$
parallel merge
(c) One-loop triangle
λ_2^{red}
2
3
y_{24}^{red}
4
λ_4^{red}
triangle integral
(4.3)
(d) Final edge
3
4
0 loops